

Agentic Quant Lab

From AI-generated hypotheses to
evidence-backed trading strategies

Agents research.
Deterministic systems decide.



VISION → DISCIPLINED RESEARCH

AI can generate trades. Can we trust them?

TYPICAL AI TRADING BOT

LLM

“Bullish”

BUY

AGENTIC QUANT
LAB

Hypothesis

Evidence

Falsify

Risk gate

Trade*

Hallucinated reasoning

Backtest overfitting

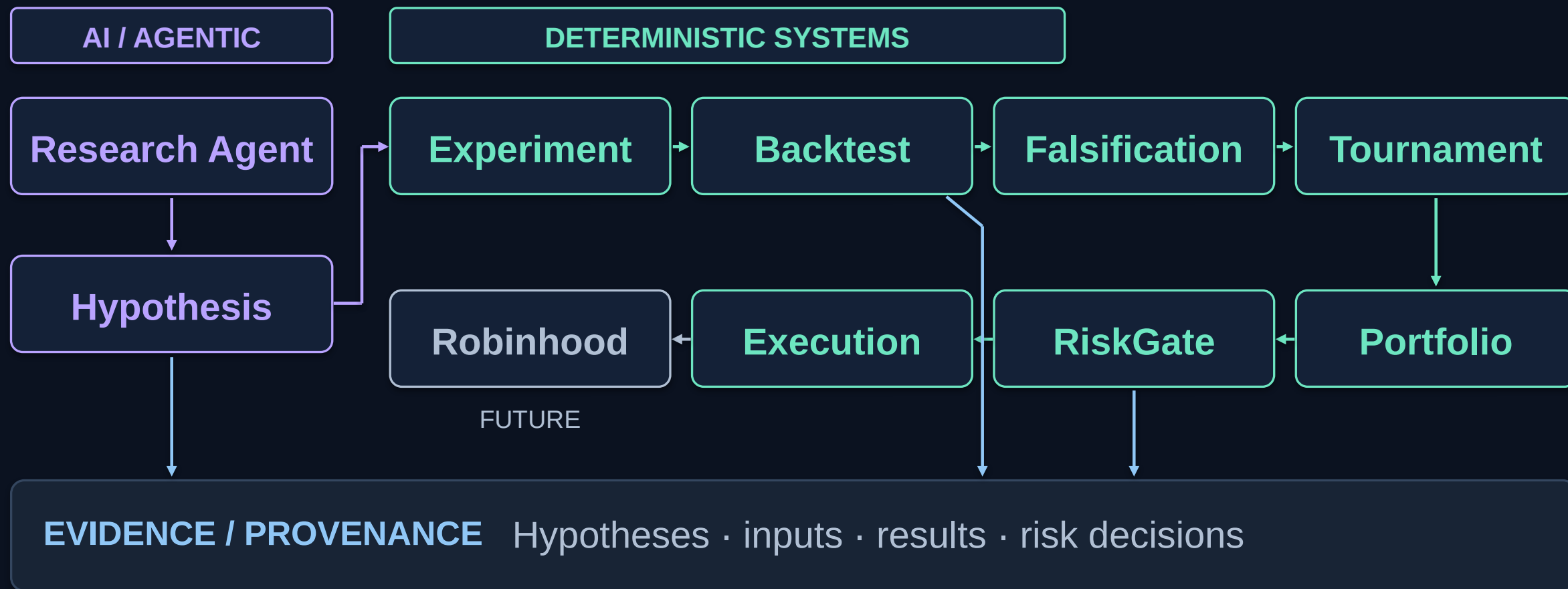
Look-ahead / bad data

Uncontrolled execution

The hard problem is rejecting bad ideas.

*Future only. Today ends at dry run.

An autonomous research loop



Target architecture. Scripted agent today; demo receipts ≠ Phase 0 ledger.

The agent has to prove itself wrong

NULL CONTROL

0

No edge?
No alpha.

PLANTED SIGNAL

+

Known edge?
Recover it.

LEAKAGE TRAP

t+1

Future data?
Reject it.

Costs

Out-of-sample

Regime tests

Data provenance

Kill the idea. Don't prompt-engineer a rescue.

Research requirements, not a claim that the demo validates market alpha.

The system is designed to say NO

ETF Trend	RESEARCH MORE	Weak versus static de-risking
Stock Momentum	DATA BLOCKED	Historical universe + delistings
Momentum + Quality	DATA BLOCKED	Stock history + PIT fundamentals
PEAD / Earnings	DEFER	Consensus blocked; price route gated
ETF Relative Momentum	NOT YET TESTED	Frozen protocol; runner DNS blocked

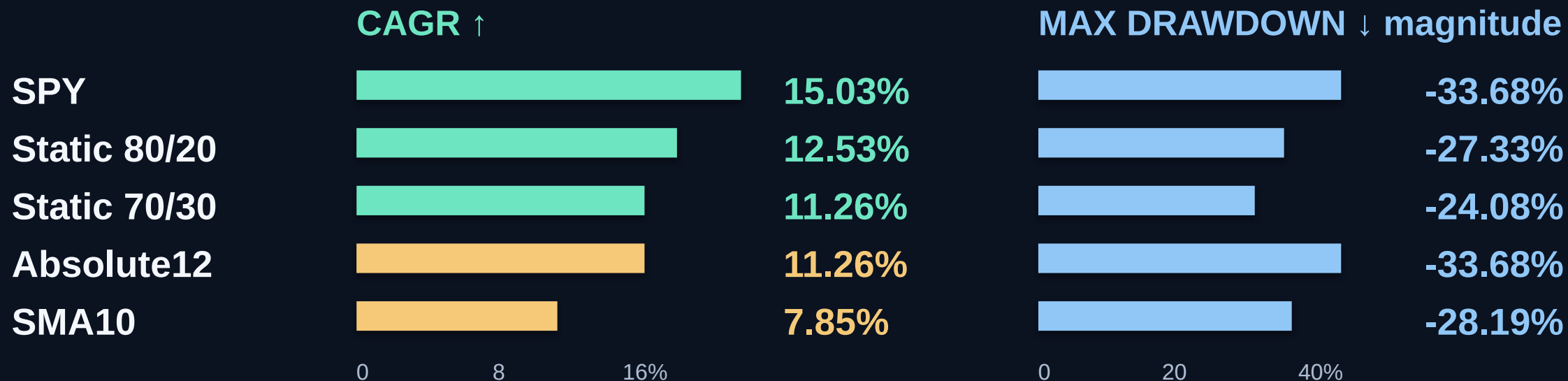
Missing evidence is not an economic rejection.

Phase 0: historical evidence acceptance recorded. No production strategy claimed.

Trend looked promising—until we tested it

REAL RESEARCH · EXPLORATORY / NAV-BASED

2016-01-04 → 2026-07-31 · SPY / BIL · 20 bps one-way costs



Absolute12: no worst-drawdown benefit

SMA10: COVID protection, then missed rebound

Losing defensive cycles: Absolute12 4/4 · SMA10 9/11

Rules stayed frozen. No optimization around the failure.

One command. Research to dry run.

```
$ uv run aql demo
```

Hypothesis → Experiment → Backtest

✓ Controls → Tournament → Portfolio

✓ RiskGate → DRY RUN

LIVE TRADING: OFF

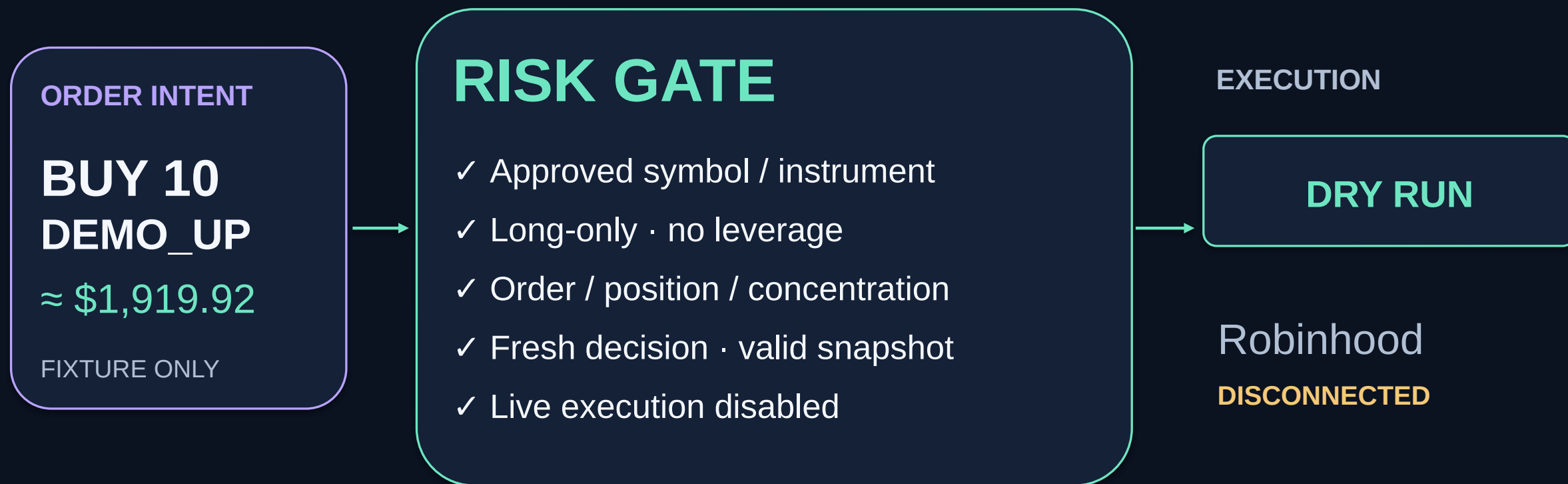
DATA: DEMO FIXTURE

EXECUTION: DRY RUN

Scripted agent. No LLM call. No market-data API.

Prepared offline: `uv run --offline --no-sync aql demo`

Agents never get direct control of capital



Agents propose. RiskGate authorizes a dry run.

One BUY / supplied demo state. No broker reconciliation.

Future design: broker credentials stay outside research agents.

Where AI actually adds value

AGENT / TARGET CAPABILITIES

Read research · form hypotheses
Interpret earnings / guidance
Investigate failure regimes
Propose the next experiment

DETERMINISTIC CODE

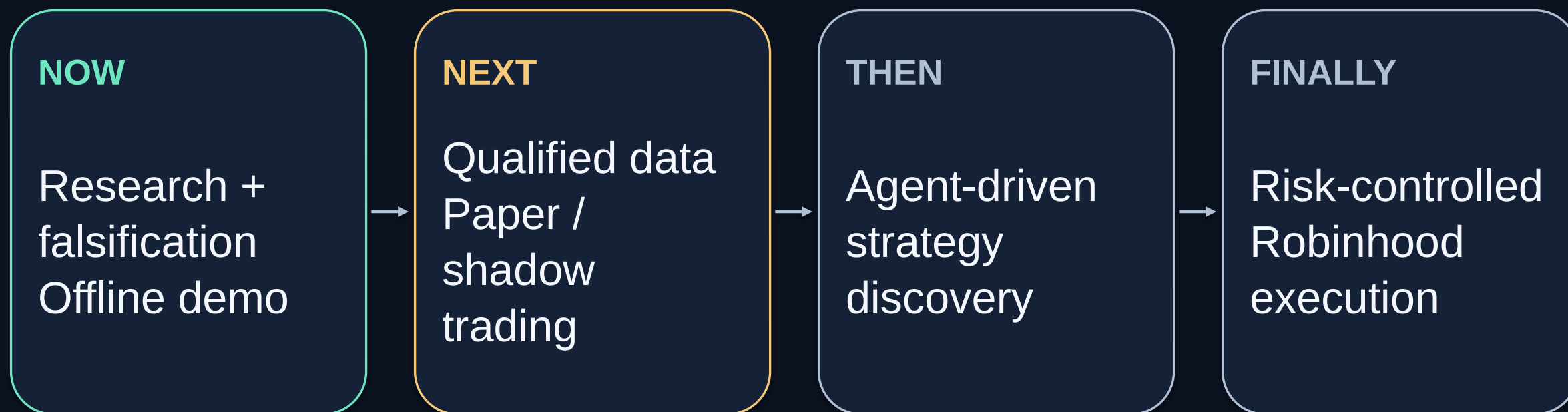
Returns · rankings · backtests
Costs · portfolio mathematics
Scientific controls · risk limits
Order validation · execution

Use AI for reasoning.

Use code for truth.

“Truth” means reproducible calculations—not certainty about markets.

From research lab to autonomous trading system



Live execution is intentionally disabled today.

Promotion requires evidence—not a better story.

The goal isn't an AI that trades more.

**It's an AI that knows
when NOT to trade.**

RESEARCH

EVIDENCE

FALSIFICATION

RISK

Agentic Quant Lab

github.com/BruceGK/agentic-quant-lab